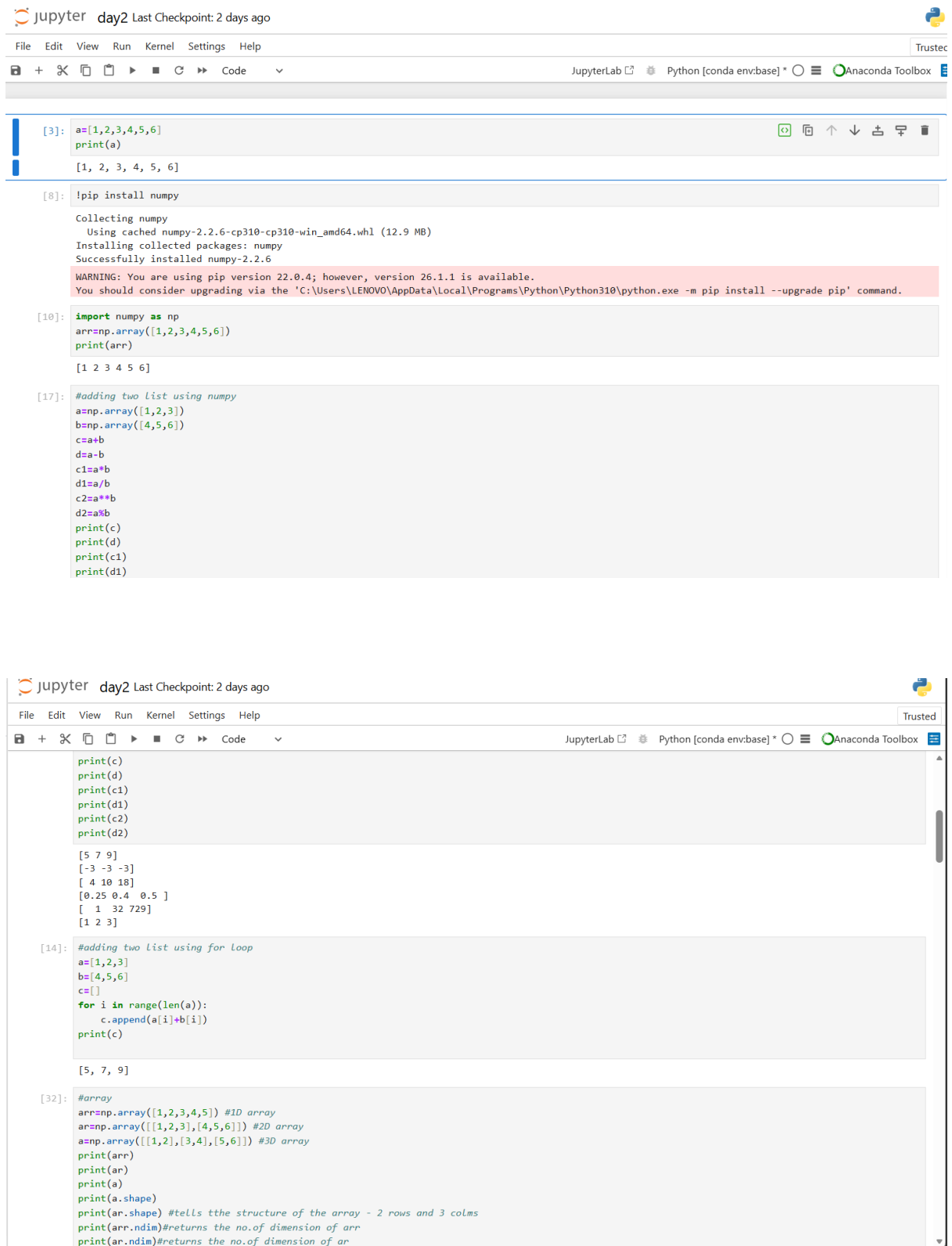


Day -2



JupyterLab day2 Last Checkpoint: 2 days ago

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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[3]: a=[1,2,3,4,5,6]
     print(a)

[1, 2, 3, 4, 5, 6]
```

```
[8]: !pip install numpy

Collecting numpy
  Using cached numpy-2.2.6-cp310-cp310-win_amd64.whl (12.9 MB)
Installing collected packages: numpy
Successfully installed numpy-2.2.6

WARNING: You are using pip version 22.0.4; however, version 26.1.1 is available.
You should consider upgrading via the 'C:\Users\LENOVO\AppData\Local\Programs\Python\Python310\python.exe -m pip install --upgrade pip' command.
```

```
[10]: import numpy as np
      arr=np.array([1,2,3,4,5,6])
      print(arr)

[1 2 3 4 5 6]
```

```
[17]: #adding two List using numpy
      a=np.array([1,2,3])
      b=np.array([4,5,6])
      c=a+b
      d=a-b
      c1=a*b
      d1=a/b
      c2=a**b
      d2=a%b
      print(c)
      print(d)
      print(c1)
      print(d1)
```

```
print(c)
print(d)
print(c1)
print(d1)
print(c2)
print(d2)

[5 7 9]
[-3 -3 -3]
[ 4 10 18]
[0.25 0.4 0.5 ]
[ 1 32 729]
[1 2 3]
```

```
[14]: #adding two List using for loop
      a=[1,2,3]
      b=[4,5,6]
      c=[]
      for i in range(len(a)):
          c.append(a[i]+b[i])
      print(c)

[5, 7, 9]
```

```
[32]: #array
      arr=np.array([1,2,3,4,5]) #1D array
      ar=np.array([1,2,3],[4,5,6]) #2D array
      a=np.array([[1,2],[3,4],[5,6]]) #3D array
      print(arr)
      print(ar)
      print(a)
      print(a.shape)
      print(ar.shape) #tells tthe structure of the array - 2 rows and 3 cols
      print(arr.ndim)#returns the no.of dimension of arr
      print(ar.ndim)#returns the no.of dimension of ar
```

```
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print(arr.size)#returns size of arr
print(ar.size) #returns size of ar
print(a.size) #returns size of a
print(a.dtype) #returns the datatype

[1 2 3 4 5]
[[1 2 3]
 [4 5 6]]
[[1 2]
 [3 4]
 [5 6]]
(3, 2)
(2, 3)
1
2
2
5
6
6
int64

[28]: import numpy as np
a=np.array([1,2,3,4,5])
b=np.zeros(5)
print(a)
print(b)
c=np.zeros((2,2)) #filled 1d array with zeros
print(c)
d=np.ones((3)) #filled 3d array with ones
print(d)
e=np.ones((2,3)) #filled 2d array with ones
print(e)

[1 2 3 4 5]
[0. 0. 0. 0. 0.]
[[0. 0.]
```

```
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[1. 1. 1.]
[[1. 1. 1.]
 [1. 1. 1.]]

[43]: import numpy as np
a=np.array([1,2,3,4,5])
b=np.zeros(5)
print(a)
print(b)
print(np.arange(1,11,3))
print(np.linspace(0,20,6))
print(np.random.rand(8)) #generates any random value within the given limit value
print(np.random.randint(1,100,6)) #gives random values from 1-100 .. (6 random numbers )
print(np.random.randn(5))

[1 2 3 4 5]
[0. 0. 0. 0. 0.]
[ 1  4  7 10]
[ 0.  4.  8. 12. 16. 20.]
[0.84715553 0.79957922 0.08489474 0.46728807 0.98064297 0.51611982
 0.56501987 0.02504164]
[12 87 54 10 51 39]
[ 0.11800525 -2.08500485 -1.13449555 -0.87177074  0.33141126]

[45]: import numpy as np
arr = np.array([10, 20, 30, 40, 50])
# Single element
arr[0]    # 10 - first element
arr[2]    # 30 - third element
arr[-1]   # 50 - last element
arr[-2]   # 40 - second from Last
print(arr[0])
print(arr[-1])

10
50
```

```
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[ ]: # *Slicing in numpy*
arr=np.array([10,20,30,40,50])
print(arr[1:4])
print(arr[0:3])
print(arr[2:])
print(arr[:3])
print(arr[:])
print(arr[::2])
print(arr[1::2])
print(arr[::-1])

[56]: arr = np.array([10, 20, 30, 40, 50])

# Condition
print(arr > 25)          # [False False True True True]

# Filter
print(arr[arr > 25])      # [30 40 50]
print(arr[arr % 2 == 0])  # [10 20 30 40 50] - even
print(arr[arr < 30])      # [10 20]

# Multiple conditions
print(arr[(arr > 15) & (arr < 45)]) # [20 30 40]
print(arr[(arr < 15) | (arr > 45)]) # [10 50]

# 2D Boolean
arr2d = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr2d[arr2d > 5])      # [6 7 8 9] - 1D result!

[False False True True True]
[30 40 50]
[10 20 30 40 50]
[10 20]
-
```

```
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```

Mathematial operation in numpy

```
[58]: import numpy as np
a=np.array([1,2,3])
print(a+10)
b=np.array([[1,2,3],[4,5,6]])
print(b+10)
print(b+a)

[11 12 13]
[[11 12 13]
 [14 15 16]]
[[2 4 6]
 [5 7 9]]

[63]: import numpy as np
a=np.array([[1,2],
           [3,4]])
b=np.array([[4,3],[2,1]])
print(a*b)
print(np.dot(a,b))
print(a@b)
print(np.matmul(a,b))
print(a.T)

[[4 6]
 [6 4]]
[[ 8  5]
 [20 13]]
[[ 8  5]
 [20 13]]
[[ 8  5]
 [20 13]]
[[1 3]
```

Statistical function

```
[66]: import numpy as np
a=np.array([10,20,30,40,50])
print(np.mean(a))
print(np.median(a))
print(np.std(a))
print(np.max(a))
print(np.min(a))
print(np.percentile(a,25))
print(np.percentile(a,50))
print(np.percentile(a,75))
print(np.var(a))
print(np.argmax(a))
print(np.argmin(a))
b=np.array([1,2,3,4,5])
c=np.array([10,2,8,5,19])
print(np.corrcoef(b,c))

30.0
30.0
14.142135623730951
50
10
20.0
30.0
40.0
200.0
0
4
[[1.         0.51418721]
 [0.51418721 1.        ]]
```

workout question

create 2D numpy array of studentt scores; compute class abverage,highest scorer, normalize score to 0-1 range

```
[68]: import numpy as np

# --- Step 1: Create 2D Array ---
# Rows = Students, Columns = Subjects
# Subjects: Math, Science, English, History

scores = np.array([
    [85, 92, 78, 90], # Alice
    [76, 65, 88, 72], # Bob
    [91, 89, 94, 87], # Clara
    [60, 70, 65, 58], # David
    [88, 95, 82, 91], # Eva
])

print("Score Array:")
print(scores)
print("Shape:", scores.shape) # (5, 4)
avg=np.mean(arr)
print(avg)
high=np.sum(arr,axis=1)
print(high)
min=np.min(high)
print(min)
max=np.max(high)
print(max)
total=np.sum(arr)
norm=(arr-arr.min())/arr.max()-min()
```

```
# --- Step 1: Create 2D Array ---  
# Rows = Students, Columns = Subjects  
# Subjects: Math, Science, English, History
```

```
scores = np.array([  
    [85, 92, 78, 90], # Alice  
    [76, 65, 88, 72], # Bob  
    [91, 89, 94, 87], # Clara  
    [60, 70, 65, 58], # David  
    [88, 95, 82, 91], # Eva  
)
```

```
print("Score Array:")  
print(scores)  
print("Shape:", scores.shape) # (5, 4)  
avg=np.mean(arr)  
print(avg)  
high=np.sum(arr,axis=1)  
print(high)  
min=np.min(high)  
print(min)  
max=np.max(high)  
print(max)  
total=np.sum(arr)  
norm=(arr-arr.min())/arr.max()-min()  
print(norm)
```

```
Score Array:  
[[85 92 78 90]  
 [76 65 88 72]  
 [91 89 94 87]  
 [60 70 65 58]  
 [88 95 82 91]]  
Shape: (5, 4)  
30.0
```